AI101 MINI-PROJECT · TEAM 66

TELCO CUSTOMER CHURN PREDICTION

Using Logistic Regression

GITHUB REPO

github.com/Siddharth-122008/telecom-customer-churn

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PROBLEM AND DATA

WHAT AND WHY

Predict which telecom customers will leave (churn), so the company can act before they go. Keeping a customer costs less than finding a new one.

DATASET SOURCE

Kaggle: Telco Customer Churn. Inputs include tenure, Contract, MonthlyCharges, InternetService and other service columns.

7,043

customers (rows)

21

columns, including the target Churn

26.5%

churned (1,869 of 7,043)

APPROACH

DATA · 1

LOAD CSV

7,043 customers, 21 columns

DATA · 2

CLEAN

Drop customerID, encode Churn as 0 and 1

DATA · 3

ENCODE

12 features become 21 numeric columns

MODEL · 4

SPLIT

80% train (5,634), 20% test (1,409)

MODEL · 5

TRAIN

StandardScaler, then Logistic Regression

OUTPUT · 6

EVALUATE

Accuracy, precision, recall, confusion matrix

LIBRARIES pandas · numpy · matplotlib · seaborn · scikit-learn

RESULTS

KEY TAKEAWAY

Month-to-month customers churn far more often than customers on one- or two-year contracts.

Contract length is one of the clearest signals in this dataset.

CONFUSION MATRIX

Predicted: Stayed (left), Churned (right)

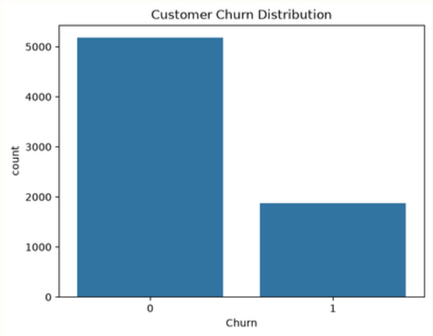
Actual Stayed	921	114
Actual Churned	174	200

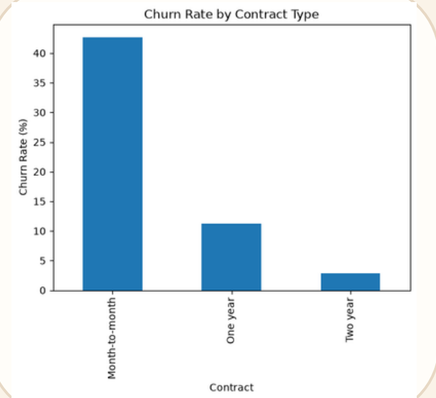
Finds 200 of 374 churners.

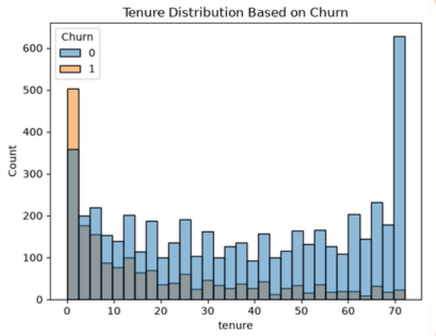
KEY NUMBERS

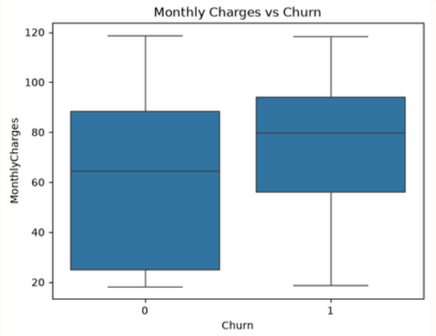
1,409 test customers · churn class

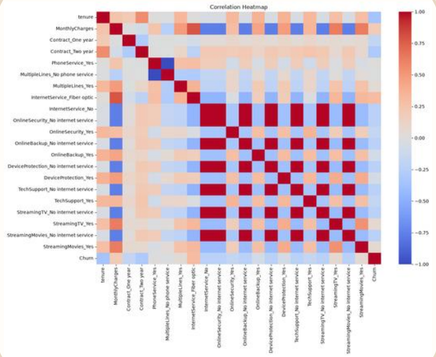
Accuracy	79.6%
Precision	63.7%
Recall	53.5%
F1 score	58.1%











LEARNINGS

WHAT WORKED

Contract type and tenure are strong signals: short contracts and new customers leave most. Accuracy reached 79.6%.

WHAT DIDN'T

Recall is only 53.5%, so the model misses almost half of the churners. MonthlyCharges overlaps with the service columns, so its coefficient is hard to read.

WHO DID WHAT

Siddharth: data loading, README

Kinjal: analysis plots, slides

Janhvi: model, run steps

Amandeep: evaluation, video

AI-USE DISCLOSURE

ChatGPT: plotting the correlation heatmap and refining the README.

Claude: debugging code, understanding errors, interpreting model , results and to improve language of ppt

All suggestions were checked by running and reviewing the code and comparing results with the notebook outputs.